

AOGS REVIEW ARTICLE

Pharmaceutical and clinical development of a novel progesterone formulation

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Key words

Progesterone, β -cyclodextrins, luteal support, *in vitro* fertilization, assisted reproductive technology, pregnancy

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Conflict of interest

The author is an employee of IBSA Institut Biochimique SA.

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Abstract

Progesterone plays an essential role in reproductive events. Its use for luteal support in patients undergoing infertility treatment is an established practice. The different routes used to administer progesterone impact on its efficacy in luteal support: oral administration has been shown to be ineffective due to an extensive first-pass metabolism in the liver; vaginal application has a good efficacy but has drawbacks such as vaginal leakage, irritation, discomfort and uncertainty about the real dose adsorbed; finally, intramuscular administration ensures a precise dosage but can be extremely painful with, in some cases, formation of sterile abscesses. A new progesterone preparation is now available in several European and extra-European countries that combines the precise dosage of the injectable formulation with the comfort of a well-tolerated subcutaneous self-administration. The pharmacokinetic and pharmacodynamic properties of this new product are reviewed here, together with the clinical evidence obtained in two multicenter randomized clinical trials.

Abbreviations: ART, assisted reproductive technology; AUC_{0-t} , area under the curve from time 0 to the last observed concentration time t ; $AUC_{0-\infty}$, area under the curve extrapolated to infinity; AUC_{ss} , area under the curve at steady state; C_{max} , maximum serum concentration; C_{ssmax} , maximum serum concentration at steady state; C_{ssmin} , minimum serum concentration at steady state; E2, estradiol; i.m., intramuscular; IVF, *in vitro* fertilization; OC, oral contraceptive; s.c., subcutaneous; T_{max} , serum peak concentration time; T_{ssmax} , serum peak concentration time at steady state; β -CD, β -cyclodextrin; β -hCG, beta human chorionic gonadotropin.

Introduction

The essential role of progesterone in reproductive events associated with the establishment and maintenance of pregnancy has already been discussed in the previous sections.

The luteal phase has been found to be defective in virtually all of the stimulation protocols used for *in vitro* fertilization (IVF). The mechanisms most likely responsible for the luteal phase defect in assisted reproductive technology (ART) include the supra-physiological levels of estradiol, decreased output of luteinizing hormone, and disruption of the corpus luteum through follicular fluid aspiration (1).

Key Message

Subcutaneous progesterone given at a daily dose of 25 mg is safe and effective in supporting the luteal phase in *in vitro* fertilization patients. The option of administering progesterone subcutaneously for luteal phase support in assisted reproductive technology will broaden the spectrum of available treatments, an advantage for women needing sustained luteal phase support or disliking vaginal treatments for cultural, personal or medical reasons.

The different routes that can be used to administer progesterone impact on its efficacy in luteal support (2,3). Oral administration has been associated with lesser efficacy (lower pregnancy rates) when used for luteal support in ART. This is due to an extensive first-pass metabolism in the liver, which leaves too little intact progesterone reaching the peripheral circulation downstream from the liver (4–6). Vaginal application permits an adequate and targeted drug delivery to the uterus thanks to the “first uterine pass effect” (7), and it is easily administered by the patient. A major drawback with this route is the vaginal leakage, which implies that the real dosage adsorbed cannot be assured. The use of vaginal progesterone is also associated with such discomfort as genital itching, genital irritation, vaginal discharge, genital yeast infection (8,9), and difficult or painful intercourse (9).

The parenteral administration of progesterone has been for a long time the only valid alternative. Although progesterone is readily soluble in oil and the parenteral administration showed a trend for enhanced efficacy when considering rates of ongoing pregnancy and live birth (10), these progesterone-in-oil preparations require an intramuscular (i.m.) administration, which is a source of significant pain and discomfort, including local soreness, sterile abscess formation, severe inflammatory and allergic reactions (11–15) due to the local tissue disruption caused by the oil (10,16). It may take more than one week to fully recover from these side-effects because the half-life of the oil vehicle in muscle is typically 2 weeks.

Subcutaneous administration has always been considered problematic due the poor hydrosolubility of the hormone, which renders the development of an aqueous solution difficult. IBSA (Institut Biochimique SA) has discovered a way of enhancing the hydrosolubility of progesterone by the addition of β -cyclodextrin (β -CD), a by-product of starch obtained by enzymatic degradation. β -CD contains six to eight glucose molecules joined in a ring by $\alpha 1 \rightarrow 4$ glycosidic bonds. Such compounds have been extensively used in the pharmaceutical industry to solubilize non-polar compounds and to improve the absorption and distribution of drugs. By using a complex of progesterone and β -CD, IBSA has developed a new aqueous formulation of progesterone (Prolutex[®]) dispensed in 25 mg vials that can be administered by subcutaneous (s.c.) injection (17). The advantages of using aqueous Prolutex[®] instead of the other progesterone-in-oil preparations mainly stem from its particular pharmacokinetic profile and from its more comfortable administration.

The aim of this article is to review the pharmacokinetic and pharmacodynamic properties of a new progesterone product, together with the clinical evidence obtained in

two multicenter randomized clinical trials, providing evidence of its safety and efficacy as an alternative treatment for luteal phase support in IVF.

Overview of clinical pharmacokinetics

Three studies have been conducted to evaluate the pharmacokinetic profile of Prolutex[®] (18). A comparative pharmacokinetic study was carried out by administering Prolutex[®] either i.m. or s.c. at a dose of 100 mg of progesterone (2×50 mg vials) or as an oily reference formulation of progesterone 100 mg (one 100 mg vial of Prontogest[®], IBSA Farmaceutici Italia). The study was conducted as a single-dose, randomized, three-way crossover study; the crossover design was chosen to minimize intra-subject variability. Twenty-four postmenopausal healthy volunteers (amenorrhoeic naturally for one year at least, or due to surgical menopause), aged between 40 and 70 years, were enrolled. Post-menopausal women have by definition low endogenous progesterone levels. Each woman received a single 100 mg dose of progesterone, administered during three consecutive study periods, separated by a 14-day wash-out interval. The three formulations were Prolutex[®] i.m. and s.c., and Prontogest[®] i.m. The pharmacokinetic profiles after administration of a single 100 mg dose of progesterone aqueous formulation by i.m. and s.c. injection and oily i.m. formulation are depicted in Figure 1. Serum progesterone was measured at pre-dose and 0.25, 0.5, 1, 2, 4, 6, 8, 10, 12, 16, 24, 36, 48, 72 and 96 h for the test and 0.5, 1, 2, 4, 6, 8, 10, 12, 16, 24, 36, 48, 72, 96 and 120 h for the reference. Pharmacokinetic analyses were performed according to a non-compartmental kinetic model, and the following parameters were calculated: maximum serum concentration ($C_{\max}$), serum peak concentration time ($T_{\max}$), half-life ($t_{1/2}$), AUC_{0-t} , $AUC_{0-\infty}$ (AUC, area under the curve), on the basis of the measured progesterone serum concentrations (Table 1).

The three treatments showed significantly different treatment effects for $C_{\max}$, but a subject effect was only noted in Prontogest[®] i.m. vs. Prolutex[®] s.c. When the AUC was calculated there was no treatment effect difference between Prontogest i.m. and Prolutex i.m., or Prolutex[®] i.m. vs. s.c. However, the AUC for Prontogest i.m. vs. Prolutex s.c. was borderline ($p = 0.05$). For all the comparisons the equivalence test at a 90% confidence interval and level of significance set at 5% applied to $C_{\max}$ could not demonstrate bioequivalence, whereas bioequivalence was always satisfied for AUC_{0-t} (reference confidence interval 0.8–1.25).

In conclusion, Prolutex i.m. and s.c. aqueous formulations gave rise to $C_{\max}$ values that were about four and

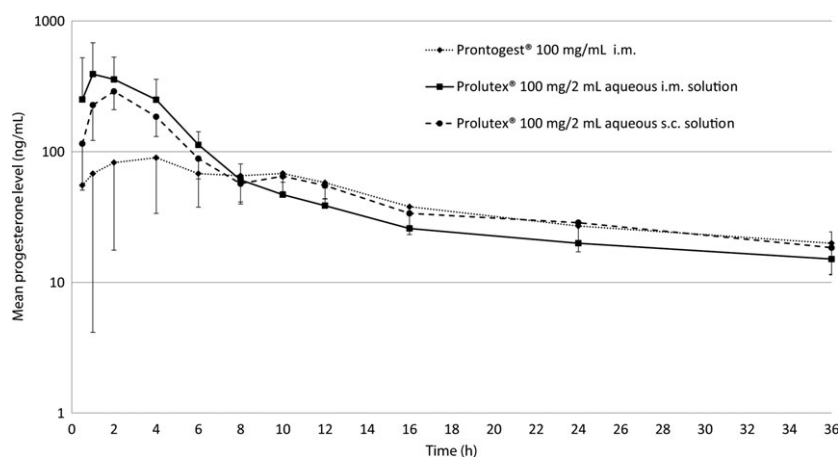


Figure 1. Mean (+SD) serum progesterone concentrations (ng/mL) vs. time profile after single 100 mg administration of progesterone aqueous formulation by intramuscular and subcutaneous injection and oily intramuscular formulation. Baseline corrected. Semi-logarithmic scale. $n = 24$.

Table 1. Main progesterone baseline-corrected pharmacokinetic parameters (mean $\pm$ SD) calculated for the three formulations under study: (A) Prontogest®, 100 mg progesterone in oily solution i.m. (B) Prolutex® 2 $\times$ 50 mg progesterone vials in aqueous i.m. (C) Prolutex 2 $\times$ 50 mg progesterone vials in aqueous solution s.c. Baseline subtracted. ($n = 24$).

Pharmacokinetic parameter	Prontogest (A)	Prolutex i.m. (B)	Prolutex s.c. (C)
C_{max} (ng/mL)	112.9 $\pm$ 57.2	440 $\pm$ 276.2	300 $\pm$ 82.6
T_{max} (h)	6.67 $\pm$ 3.79	0.88 $\pm$ 0.53	0.92 $\pm$ 0.19
AUC_{0-t} (ng/mL/h)	2049 $\pm$ 259	1902 $\pm$ 287	1856 $\pm$ 295
$AUC_{0-\infty}$ (ng/mL/h)	2097 $\pm$ 263	1919 $\pm$ 289	1885 $\pm$ 290
$t_{1/2}$ (h)	22.3 $\pm$ 11.6	14.3 $\pm$ 4.6	17.2 $\pm$ 11.5

AUC_{0-t} , area under the curve from time 0 to the last observed concentration time t ; $AUC_{0-\infty}$, area under the curve extrapolated to infinity; C_{max} , maximum serum concentration; i.m., intramuscular; s.c., subcutaneous; T_{max} , serum peak concentration time, $t_{1/2}$, half-life.

three times higher compared with i.m. injections of progesterone-in-oil. C_{max} values were reached approximately nine and eight times faster, respectively. As expected, the total progesterone bioavailability did not differ statistically among formulations. A statistical difference was noted between the aqueous and oily progesterone preparations for T_{max} . However, no difference was noted in T_{max} when the aqueous formulation (Prolutex®) was administered i.m. or s.c.

The second study investigated the dose linearity of Prolutex® within the dose ranges commonly used in clinical practice in 12 postmenopausal healthy volunteers. Women randomly received a single dose by s.c. injection of Prolutex 25, 50 or 100 mg. This was a three-way cross-over study, with a 14-day washout between medications.

The pharmacokinetic profiles after a single s.c. administration of 25, 50 or 100 mg of progesterone are depicted in Figure 2. Blood samples for progesterone were collected at the time-points reported above for the test product. The measured serum concentrations allowed the calculation of the PK profile of Prolutex deriving from the s.c. administration of 25, 50 and 100 mg (Table 2). The findings of the study indicate that the serum pharmacokinetic of Prolutex follow a dose-linear relation in terms of C_{max} and AUC, when administered as single s.c. doses of 25, 50 and 100 mg to postmenopausal women volunteers.

In the third study, the pharmacokinetic profile of steady-state progesterone following two treatments (25 and 50 mg/day) of Prolutex® was investigated after a repeated administration for 11 days in 24 premenopausal healthy volunteers (Figure 3). Progesterone concentrations were measured every day before the administration of the product (i.e. daily pre-dose assessment) and on day 11 at the time-points listed above, up to 72 h post-dose. Following the last administration, pharmacokinetic parameters at steady-state were calculated after baseline subtraction (results are reported in Table 3). Peak concentrations (C_{ssmax}) were reached 1.0 $\pm$ 0.4 h and 0.9 $\pm$ 0.4 h after administration of Prolutex 25 and 50 mg, respectively. Afterwards, serum progesterone decreased with a mono-exponential decay, and serum progesterone returned to baseline concentrations 48–72 h after the last dose. Steady-state minimum progesterone concentration (C_{ssmin}) was 3.4 $\pm$ 1.0 and 6.6 $\pm$ 2.0 ng/mL for the 25 and 50 mg doses, respectively. The kinetic results demonstrated the linearity of the progesterone pharmacokinetic within the range of doses tested. In addition, the ratio of the AUC at steady state (AUC_{ss})

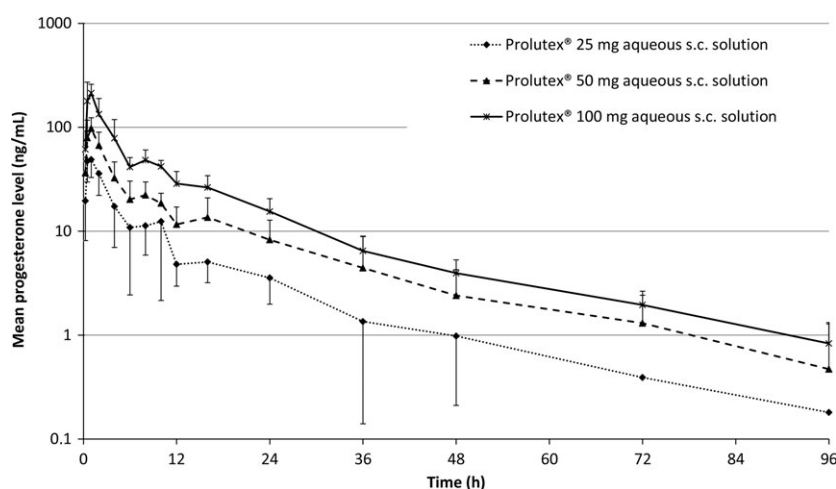


Figure 2. Mean ($\pm$ SD) serum progesterone concentrations (ng/mL) vs. time profile after single subcutaneous administration of progesterone aqueous formulation 25, 50 and 100 mg. Baseline corrected. Semi-logarithmic scale. $n = 12$.

Table 2. Main progesterone baseline-corrected pharmacokinetic parameters (mean $\pm$ sd) calculated after subcutaneous administration of a Prolutex® single dose of 25, 50 and 100 mg. $n = 12$.

Pharmacokinetic parameter	25 mg	50 mg	100 mg
C_{max} (ng/mL)	57.8 $\pm$ 13.6	103 $\pm$ 24.5	235 $\pm$ 62.6
T_{max} (h)	0.92 $\pm$ 0.42	0.92 $\pm$ 0.42	0.92 $\pm$ 0.42
AUC_{0-t} (ng/mL/h)	338 $\pm$ 91.6	729 $\pm$ 228	1466 $\pm$ 213
$AUC_{0-\infty}$ (ng/mL/h)	349 $\pm$ 91.1	746 $\pm$ 250	1490 $\pm$ 213
$t_{1/2}$ (h)	13.1 $\pm$ 7.1	17.2 $\pm$ 5.1	17.6 $\pm$ 5.8

AUC_{0-t} , area under the curve from time 0 to the last observed concentration time t ; $AUC_{0-\infty}$, area under the curve extrapolated to infinity; C_{max} , maximum serum concentration; T_{max} , serum peak concentration time; $t_{1/2}$, half-life.

after dose correction demonstrated similar dose-normalized bioavailability for the two formulations. After s.c. administration, doses of 25–50 mg result in serum progesterone concentrations that are comparable to those achieved in the luteal phase of a spontaneously occurring menstrual cycle (i.e. 2–20 ng/mL) (19).

Overview of clinical efficacy

In patients treated by ART, it is difficult to determine precisely the extent to which progesterone is needed, let alone determine the most efficient and minimal effective doses. This complexity stems from the fact that in ART, progesterone is produced not by one but rather by the multitude of corpora lutea triggered by the controlled ovarian hyperstimulation. There is therefore an inherent difficulty in determining the exact amount of proges-

terone needed in each individual case because controlled ovarian hyperstimulation responses are inherently variable.

A markedly different situation prevails in women whose ovaries have prematurely failed or who have been down-regulated with gonadotrophin releasing-hormone analogue treatments and who need exogenous hormones [estradiol (E2) and progesterone] for their endometrium to be receptive to embryos obtained from donated oocytes. In the recipients of donor egg IVF, there is no endogenous production of progesterone. Hence, the priming of endometrial receptivity depends solely on the exogenous hormones administered in an effort to duplicate ovarian function. Yet, the hormonal regimens designed for women whose ovaries are inactive actually provide optimal endometrial receptivity, which is demonstrated by the notoriously high pregnancy rates of donor egg IVF. Consequently, the E2 and progesterone cycles conceived for donor egg ART have become an ideal and universally recognized model (3,20,21) for studying the efficacy of hormonal treatments at priming endometrial receptivity.

Therefore, prior to undertaking a Phase III safety and efficacy trial, a dose-ranging study was conducted to determine the minimum dose of Prolutex® that could effectively provide luteal phase support in ART. This was undertaken using the donor egg ART model discussed above. This approach challenges the progesterone preparation in an environment free of endogenous hormone, as previously done for other progesterone preparations (22,23).

The dose-ranging study was conducted in 24 normal volunteer women from 20 to 45 years of age (24). All

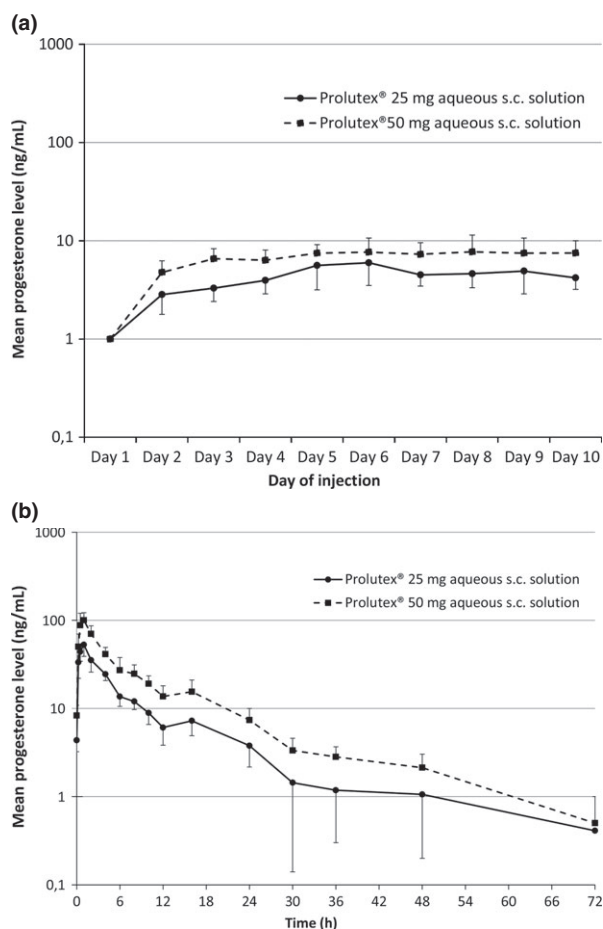


Figure 3. (a) Mean (+SD) pre-dose serum progesterone concentrations (ng/mL) on each of the 10 days of subcutaneous administration of progesterone aqueous formulation 25 and 50 mg. Baseline corrected. Semi-logarithmic scale. $n = 12$ in each group. (b) Mean (+SD) serum progesterone concentrations (ng/mL) vs. time profile after the 11th day of subcutaneous administration of progesterone aqueous formulation 25 and 50 mg. Baseline corrected. Semi-logarithmic scale. $n = 12$ in each group.

women were seen for an inclusion visit on days 1–15 of their menstrual cycle or while on oral contraceptive (OC). After confirming the absence of an underlying pathology, all women received a prescription for starting or extending their use of OC (ethinyl estradiol 0.03 mg and levonorgestrel 0.125 mg). OC was taken (or extended) for 10–24 days for the purpose of synchronizing the study onset. A single injection of a depot preparation of leuprolide acetate was administered two to four days prior to stopping OC. On the second day of menses after stopping OC, estrogen therapy was initiated using transdermal E2. Twenty days later, plasma progesterone was measured to confirm the absence of endogenous production of progesterone (<1 ng/mL). Subsequently,

Table 3. Main progesterone baseline-corrected pharmacokinetic parameters (mean $\pm$ SD) at the steady-state after administration of Prolutex® 25 or 50 mg s.c. $n = 12$ for each group.

Pharmacokinetic parameter	T1: Prolutex® 25 mg	T2: Prolutex® 50 mg
C_{ssmax} (ng/mL)	53.1 ± 13.7	110 ± 20.7
C_{ssmin} (ng/mL)	3.34 ± 1.0	6.6 ± 2.0
T_{ssmax} (h)	1.0 ± 0.4	0.9 ± 0.4
AUC_{ss} (ng/mL/h)	313 ± 44.4	615 ± 93.4
$Vdss$ (L)	762 ± 205	830 ± 152
$Clss$ (L/h)	81.5 ± 12.3	83.1 ± 13.0
%PTF	383 ± 93.2	408 ± 81.0
%Swing	1657 ± 832	1740 ± 720

AUC_{ss} , area under the curve at steady state; $Clss$, clearance; C_{ssmax} , maximum serum concentration at steady state; C_{ssmin} , minimum serum concentration at steady state; PTF: peak through fluctuation; T_{ssmax} , serum peak concentration time at steady state; $Vdss$, distribution volume.

women were prospectively randomized to receive either 25 or 50 mg of Prolutex® daily in the form of self-administered subcutaneous injections. An endometrial biopsy was obtained on the 11th day of progesterone exposure. Results were analyzed in a blinded fashion by two independent pathologists who were both experts in endometrial morphology. The study protocol stipulated that divergent readings would be discussed after conducting a slide review and an adjudicating pathologist was available if agreement could not be reached. Furthermore, on the last (11th) day of progesterone administration, blood samples were drawn to determine steady state pharmacokinetics. Drug tolerability was assessed by reviewing a self-filled diary.

The size of the study groups ($n = 12$) was set according to past experience in trials looking at progesterone effects in E2 and progesterone cycles designed for donor egg IVF. This number was set according to the following objective – to obtain a predecidualized endometrium (characteristic condensation of endometrial stroma) in all interpretable biopsies.

All the treated women showed evidence of adequate estrogenization at their first visit. Of 24 biopsies performed, 12 with each progesterone dose, two did not contain tissue. All the assessable biopsies ($n = 22$) showed predecidual changes of the endometrial stroma characteristic of the late luteal phase. There were no differences between the two treatment doses with respect to any of the parameters evaluated. In other words, daily subcutaneous administration of 25 or 50 mg progesterone induces similar predecidual endometrial changes, a finding that is in agreement with earlier observations obtained with 25 mg of i.m. progesterone (25).

The dose-ranging study challenged the ability of Prolutex[®] to induce the full array of endometrial changes observed in the late luteal phase of the menstrual cycle and described as “predecidual changes”. The sought-after effect – the complete secretory transformation of the endometrium – is a unitary one beyond which no further beneficial effects can be achieved, as discussed above. These results unequivocally showed that the lower of the two doses studied, 25 mg/day, is adequate and sufficient for sustaining the luteal phase, even in the most adverse conditions, when there is no endogenous production of progesterone. The serum progesterone profile obtained with 25 mg/day corresponds to the upper end of the daily production of progesterone in the menstrual cycle (19). For this reason, 25 mg/day was considered the most appropriate dose to be evaluated in a Phase III safety and efficacy trial for its ability to sustain luteal phase in patients undergoing ART cycles.

Clinical efficacy studies

The safety and efficacy of Prolutex[®] 25 mg as luteal support for patients undergoing IVF has been tested in two prospective, open-label, randomized, controlled, parallel-group, multicenter, two-arm, non-inferiority studies (30,31). In the first study, 683 women enrolled in 13 European fertility clinics were treated with Prolutex[®] 25 mg/day, given s.c., or by vaginal gel 90 mg (Crinone[®], Merck Serono, Darmstadt, Germany). In the second study, 800 patients enrolled in eight fertility clinics in the USA were treated with Prolutex[®] 25 mg/day or with vaginal tablets 100 mg twice a day (Endometrin[®], Ferring Pharmaceuticals, Saint-Prex, Switzerland). In both studies, treatment duration was 10 weeks, a period consistent with current practice for luteal support (from oocyte

retrieval until pregnancy testing, or until the luteo-placental shift in the event of pregnancy), and efficacy was assessed objectively based on on-going pregnancy rates. The women enrolled in these studies were aged between 18 and 42 years upon starting controlled ovarian hyperstimulation. They had undertaken fewer than three prior ART cycles (IVF and related procedures). At baseline (day 2–3 of cycling) follicle-stimulating hormone was ≤ 15 IU/L and E2 < 80 pg/mL. All patients had normal uterine cavities and had at least three retrieved oocytes.

Outcome of the studies

In both studies, the two treatment groups were well matched as to demographic and infertility history. No notable differences were observed between the treatment groups for the oocyte retrieval data (number of oocytes retrieved, number of mature oocytes, number of inseminated oocytes or number of injected oocytes). There were no statistically significant or clinically meaningful differences between the treatment groups in the numbers of embryos obtained or transferred or in the women's endometrial thickness on the day of embryo transfer.

In both studies, the on-going pregnancy rates at 10 weeks were comparable in all treatment groups (Tables 4 and 5 for the European and the USA study, respectively). The non-inferiority of Prolutex[®] treatment compared with Crinone[®] and Endometrin[®] treatment in the primary efficacy endpoint of on-going pregnancy rates at 10 weeks was established, since the lower bound of the 95% confidence interval of the difference between the two proportions was greater than the non-inferiority margin of -0.10 .

With regard to the secondary end-points considered in the two studies, there were no statistically significant

Table 4. On-going pregnancy rate at 12 weeks of gestation [intention-to-treat (ITT) and per protocol (PP) analysis populations], European study.

Pregnancy rate, <i>n</i> (%)	Prolutex [®]	Crinone [®]	<i>p</i> -value	Difference vs. Crinone(95%CI)
ITT population ¹	93 (27.4)	105 (30.5)	0.40	−3.09 (−9.91%, 3.73%)
PP population ²	93 (29.2)	100 (31.2)	0.61	−2.00 (−9.12%, 5.13%)

¹*n* = 339 in the Prolutex[®] group; *n* = 344 in the Crinone[®] group.

²*n* = 319 in the Prolutex[®] group; *n* = 321 in the Crinone[®] group.

Table 5. On-going pregnancy rate at 12 weeks of gestation [intention-to-treat (ITT) and per protocol (PP) analysis populations], USA study.

On-going pregnancy rate, <i>n</i> (%)	Prolutex [®]	Endometrin [®]	<i>p</i> -value	Difference vs. Endometrin (95%CI)
ITT population ¹	163 (40.8)	173 (43.3)	0.52	−2.5 (−9.4%, 4.4%)
PP population ²	163 (41.6)	173 (44.4)	0.47	−2.8 (−9.7%, 4.2%)

¹*n* = 400 in both the Prolutex[®] and in the Endometrin[®] group.

²*n* = 392 in the Prolutex[®] group; *n* = 390 in the Endometrin[®] group.

differences in implantation rates, positive beta human chorionic gonadotropin (β -hCG) test rates, clinical pregnancy rates, biochemical pregnancy rates (defined as a pregnancy that stops growing before it is large enough to be seen on ultrasound), early spontaneous abortions (spontaneous abortions before the end of the study) rates, delivery and live birth rates between the two treatment groups (Tables 6 and 7).

Regarding local tolerability at administration site, in the European study 57% of patients in the Prolutex group experienced discomfort (irritation, pruritus or hematoma) at the injection site, mainly of mild intensity. Regarding vaginal discomfort, 50.8% of the patients in the Crinone group experienced irritation, inflammation, dryness, pruritus, discharge or pain.

In the USA study, adverse events, which could be associated with route of administration, occurred with similar frequency in each group. In the Prolutex group, 22.0% of women reported adverse events associated with injection, including injection site bruising, inflammation, edema and injection site pain. In the Endometrin group, 14.5%

reported side effects associated with vaginal administration, including genital pruritus, vaginal discharge and vaginal pain/discomfort.

Discussion

The aim of this review was to summarize the studies performed with a new progesterone preparation, which differs from the already available preparation in its dosage (25 mg/day) and method of administration (subcutaneous).

Pharmacokinetic studies have demonstrated that, when the same dosage is administered, Prolutex[®] s.c. aqueous formulation and the i.m. progesterone-in-oil formulation

Table 6. Secondary efficacy analysis results [intention-to-treat (ITT) and per protocol (PP) analysis populations], European study.

	Prolutex [®]		Crinone [®]		<i>p</i> -value*
	Mean	SD	Mean	SD	
Implantation rate (PP)	22.8	35.1	22.7	32.9	0.97
	<i>n</i>	%	<i>n</i>	%	
Positive β -hCG test rate (ITT)	134	39.5	148	43.0	0.35
Positive β -hCG Test rate (PP)	134	42.0	141	43.9	0.62
Clinical pregnancy rate (ITT)	103	30.4	113	32.9	0.49
Clinical pregnancy rate (PP)	103	32.3	108	33.6	0.72
Early spontaneous abortions ^a	14	10.4	14	9.9	0.88
Pregnancy loss rate ^b	22	6.5	18	5.2	0.48
Delivery & live birth rate (ITT)	91	26.8	103	29.9	0.37
Delivery rate and live birth rate (PP)	91	28.5	98	30.5	0.58

β -hCG, beta human chorionic gonadotropin.

ITT population: *n* = 334 in the Prolutex[®] and in the Crinone[®] group.

PP population: *n* = 319 in the Prolutex[®] group; *n* = 321 in the Crinone[®] group.

*F-test (ANOVA) for continuous variables, Fisher's exact test for categorical variables.

^aCalculated on the number of positive β -hCG tests: *n* = 134 in the Prolutex group; *n* = 141 in the Crinone group.

^bCalculated on the number of clinically pregnant women: *n* = 103 in the Prolutex group; *n* = 108 in the Crinone group.

Table 7. Secondary efficacy analysis results [intention-to-treat (ITT) and per protocol (PP) analysis populations], USA study.

Secondary efficacy parameter	Prolutex [®]		Endometrin [®]		<i>p</i> -value*
	Mean	SD	Mean	SD	
Implantation rate (PP population)	33.2	42.0	35.1	40.9	0.54
	<i>n</i>	%	<i>n</i>	%	
Positive β -hCG test rate (ITT population)	221	55.3	230	57.5	0.57
Positive β -hCG test rate (PP population)	221	56.4	230	59.0	0.47
Clinical pregnancy rate (ITT population)	167	41.8	181	45.3	0.35
Clinical pregnancy rate (PP population)	167	42.6	181	46.4	0.31
Early spontaneous abortions ^a	54	24.4	49	21.3	0.38
Loss of pregnancy total ^b	5	3.0	11	6.1	0.21
Delivery rate (ITT population)	162	40.5	170	42.5	0.62
Delivery rate (PP population)	162	41.3	170	43.6	0.56
Live birth rate (ITT population)	161	40.3	168	42.0	0.67
Live birth rate (PP population)	161	41.1	168	43.1	0.61
Take-home baby rate (ITT population)	161	40.3	166	41.5	0.77
Take-home baby rate (PP population)	161	41.1	166	42.6	0.72

β -hCG, beta human chorionic gonadotropin.

ITT population: *n* = 400 in the Prolutex[®] and the Endometrin[®] group.

PP population: *n* = 392 in the Prolutex[®] group; *n* = 390 in the Endometrin[®] group.

*F-test (ANOVA) for continuous variables, Fisher's exact test for categorical variables.

^aCalculated on the number of positive β -hCG tests: *n* = 221 in the Prolutex group; *n* = 230 in the Endometrin group.

^bCalculated on the number of clinically pregnant women: *n* = 167 in the Prolutex group; *n* = 181 in the Endometrin group.

achieved comparable AUC values even when the $C_{\max}$ and the $T_{\max}$ values were significantly different, showing more rapid bioavailability of the new product. Subsequently, a dose-ranging study revealed that the daily administration of 25 mg of Prolutex[®] duplicates the endometrial changes seen in the menstrual cycle. The two efficacy studies have also clearly shown that 25 mg/day of Prolutex[®] is sufficient to ensure the adequate luteal support needed in IVF. In fact, the non-inferiority of Prolutex[®] treatment compared with treatment with vaginal preparations (Crinone[®] and Endometrin[®]) was established for the primary efficacy endpoint of on-going pregnancy rates at 10 weeks. Moreover, no statistically significant differences between the treatment groups were reported for any of the secondary efficacy endpoints (i.e. mean implantation rate, positive β -hCG test rate, clinical pregnancy rate at four to five weeks of treatment, biochemical pregnancy rate or early spontaneous abortions). Prolutex[®] can therefore be considered a good alternative treatment in IVF.

Injectable formulations offer the advantage of providing measurable and constant circulating levels of progesterone that are assessable by the investigator who wishes to monitor the patient's needs (not possible with topical preparations due to vaginal leakage). This constitutes the main reason that has led several clinicians in Europe and the USA still to prefer using injectable progesterone-in-oil preparations in spite of this being quite painful for the patient (11,12). Prolutex[®] s.c. therefore offers the advantage of a defined daily dose leading to measurable blood levels without the painful drawbacks of i.m. oily preparations.

The administration of progesterone in injectable or vaginal form is more efficient than by the oral route, since it avoids the metabolic losses of progesterone encountered with oral administration resulting from the hepatic first-pass effect (32). In addition, the injectable forms avoid the need for higher doses that cause a fairly large number of side-effects, such as somnolence, sedation, anxiety, irritability and depression (33). Moreover, the injectable s.c. formulation avoids the discomfort caused by topical vaginal preparations, such as vaginal itching, perineal irritation, vaginal leakage and general discomfort. For example, gel accumulation in the vagina has to be manually removed, and can cause pain and discomfort during intercourse (9).

Daily administration of 25 mg of progesterone – a dose corresponding to the normal production of progesterone by the corpus luteum (19) – induces physiological levels of plasma progesterone, which is comparable to findings made in the menstrual cycle (26). Existing data with other parenteral progesterone preparations indicate that the i.m. administration of 25 mg/day provides effective

luteal support (25). Furthermore, comparative trials showed no further advantages in using higher doses (25–27). In this context, the use of both the 25 and 50 mg dose in the Phase III study was deemed to be counterproductive and likely to create more confusion than true information, as both doses would provide similar effects. Therefore, the studies were performed using only the 25 mg dose. Unfortunately, supplementation of the luteal phase in IVF started empirically, and the data are limited with regard to the best dose needed (10). A confusion as to the dose to be used is precisely the situation that prevails for all the progesterone preparations currently approved for luteal support in ART.

In the case of the micronized progesterone in capsules, trials concluded that the doses of 300 and 600 mg/day produced an equivalent physiological endometrium (21,28) and clinical pregnancy rate. In spite of this, the most commonly used daily dosage is 600 mg/day (3).

In the same way, in the attempt to compare the dose of i.m. progesterone, Check et al. (25) found no difference in clinical and ongoing pregnancy rates when comparing 25 and 100 mg of daily i.m. progesterone. Nevertheless, the most widely used dose remains 50–100 mg/day (1). A similar confusion exists in the case of the vaginal gel Crinone[®]. Crinone is approved for ART at two different dosing regimens of 90 mg once or twice a day. The dosage of the most recently approved product, Lutinus[®] (Ferring, Pharmaceuticals, Saint-Prex, Switzerland marketed in the USA under the trade name Endometrin[®]), is proposed as 100 mg tablets two or three times a day (29).

The degree of confusion as to the dose of progesterone that needs to be used has led clinicians to prescribe the highest dosage with an intent to err on the safe side. However, this practice is highly undesirable, as it is likely to create an unnecessary financial burden as well as paving the way for unnecessary hormonal exposure, which could increase the risk of imprinting genetic disorders (34,35). To avoid perpetuating the confusion that already exists regarding the dose of progesterone to be used in ART, only one dose (25 mg/day) was tested in two safety and efficacy studies, showing non-inferiority against progesterone gel 90 mg (Crinone[®]) and vaginal insert 200 mg (Endometrin[®]).

In conclusion, the studies presented in this review provide evidence that 25 mg/day Prolutex[®] is safe and effective in supporting the luteal phase in IVF patients. The option of administering progesterone subcutaneously for luteal phase support in ART will broaden the spectrum of available treatments, an advantage for women needing sustained luteal phase support (36) or disliking vaginal treatments for cultural, personal or medical reasons. The 25 mg/day dose used in the two randomized controlled trials reflects the physiologic endogenous production of

progesterone by the corpus luteum during the luteal phase of a natural menstrual cycle. Further studies aimed to test even lower doses may be of interest to confirm the lowest effective dose needed for luteal phase support in IVF. For the present, use of the lowest tested dose can be a good starting point to reduce unnecessary dosage.

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